

 SpiceCart — Full UI/UX Design & Development Blueprint

A premium spice e-commerce platform inspired by Amazon/Flipkart, exclusively for spices.

1.  BRAND IDENTITY & COLOR SYSTEM

Primary Color Palette (Spice-Inspired)

Color

Hex    Usage    WCAG Contrast

|                                                                                                    |         |                                          |                                                                                             |
|----------------------------------------------------------------------------------------------------|---------|------------------------------------------|---------------------------------------------------------------------------------------------|
| Saffron Gold      | #E2B714 | Primary brand, CTAs, highlights          |  Black   |
| Turmeric Yellow                                                                                    | #F4C430 | Secondary accents, badges, stars         |  Black   |
| Chili Red         | #C41E3A | Sale tags, errors, urgency, low stock    |  White   |
| Cardamom Green    | #2E8B57 | Organic labels, success states, in-stock |  White   |
| Cinnamon Brown    | #8B4513 | Headers, rich textures, nav              |  White   |
| Pepper Dark                                                                                        | #1A1A2E | Primary text, dark mode base             |  White   |
| Cream White                                                                                        | #FFF8E7 | Light backgrounds, cards                 |  Black   |
| Paprika Orange  | #E8590C | Warnings, offer CTAs                     |  White   |
| Mint Light                                                                                         | #4A7C59 | Freshness indicators, subtle accents     |  White |

Dark Mode Palette

Color

Hex    Usage

|                 |           |                          |
|-----------------|-----------|--------------------------|
| Dark Background | #1A1A2E   | Main background          |
| Card Dark       | #16213E   | Card surfaces            |
| Surface Dark    | #0F3460   | Elevated surfaces/modals |
| Saffron Glow    | #E2B71480 | Primary with 50% opacity |
| Text Primary    | #F5F5F5   | Main body text           |
| Text Secondary  | #A0A0B0   | Subtitles, captions      |

## Semantic Colors

### Type

Color   Hex

Success      Green   #2E8B57

Warning      Orange #E8590C

Error   Red    #C41E3A

Info    Blue    #3498DB

Organic/Badge      Mint    #4A7C59

## 2. TYPOGRAPHY SYSTEM

### Font Stack

#### Level

Font    Weight Use

Display Playfair Display      700, 800      Hero banners, splash screen

Headings      Poppins      600, 700      Section titles, card titles

Body    Inter    400, 500      Product descriptions, reviews

Price    DM Sans      700      Price tags, amounts

Badge/Tag      Space Grotesk 600      Sale badges, category pills

### Type Scale (Responsive)

Display1: 36sp / 44sp line-height (Splash, Hero)

Display2: 30sp / 38sp (Feature headers)

H1:      24sp / 32sp (Screen titles)

H2:      20sp / 28sp (Section headers)

H3:      18sp / 26sp (Card titles)

Body1:    16sp / 24sp (Primary body)

Body2: 14sp / 20sp (Secondary text)

Caption: 12sp / 16sp (Metadata, timestamps)

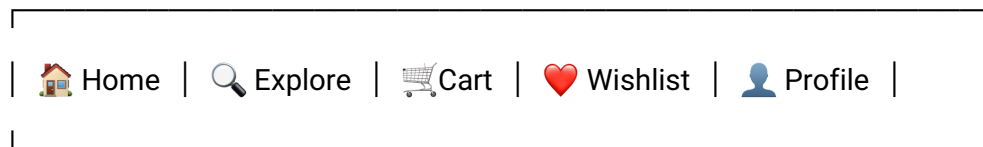
Overline: 10sp / 16sp (Labels, letter-spacing: 1.5)

Price: 22sp / bold (Current price)

Strike: 14sp / 400 / line-through (Original price)

### 3. 📱 APP ARCHITECTURE & NAVIGATION

Navigation Flow (Bottom Tab Bar)



Full Screen Map

SPLASH SCREEN



ONBOARDING (3 screens, first-time only)



LOGIN / SIGNUP

- └ Phone OTP Login
- └ Email/Password
- └ Google OAuth
- └ Guest Mode (restricted: browse only, no purchase)



HOME (Main Hub)

- └ Search Bar → Search Results Page
- └ Category Carousel
- └ └ Whole Spices

- | | — Ground Spices
- | | — Spice Blends/Masalas
- | | — Herbs & Leaves
- | | — Exotic/Rare Spices
- | | — Organic Spices
- | | — Regional Specialties
- | — Flash Deals Section
- | — Trending Products
- | — Featured Brands
- | — Recipe-Paired Spices



## EXPLORE (Full catalog)

- | — Filter & Sort Panel
- | — Grid/List Toggle
- | — Infinite Scroll Products
- | — AI Recommendations



## PRODUCT DETAIL PAGE

- | — Image Gallery (zoom, 360°)
- | — Nutrition & Origin Info
- | — Spice Heat Meter (🌶️ 1-5)
- | — Recipe Suggestions
- | — Reviews & Ratings
- | — Q&A Section
- | — Similar Products

└─ Add to Cart / Wishlist



## CART

└─ Item Management (qty, remove)

└─ Coupon/Promo Code

└─ Spice Pairing Suggestions

└─ Price Breakdown

└─ Checkout →



## CHECKOUT

└─ Address Selection/Add

└─ Delivery Scheduling

└─ Payment Methods

└─ Order Summary

└─ Order Confirmation



## ORDER TRACKING

└─ Real-time Status

└─ Map Integration

└─ Delivery ETA



## PROFILE

└─ My Orders

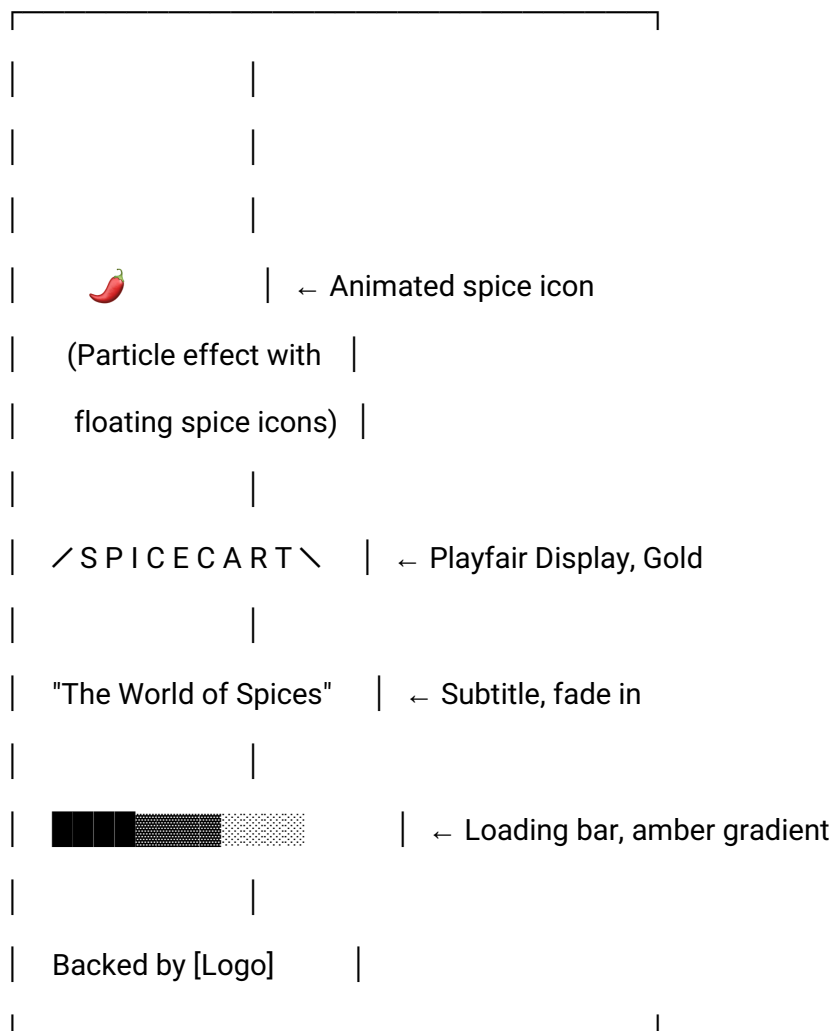
└─ Addresses

└─ Payments

- └─ Wishlist
- └─ Notifications
- └─ Settings (Theme, Lang, Region)
- └─ Help & Support
- └─ About / Logout

#### 4. 🖥️ COMPLETE SCREEN-BY-SCREEN UI DESIGN

##### Screen 1: SPLASH SCREEN (3 seconds)



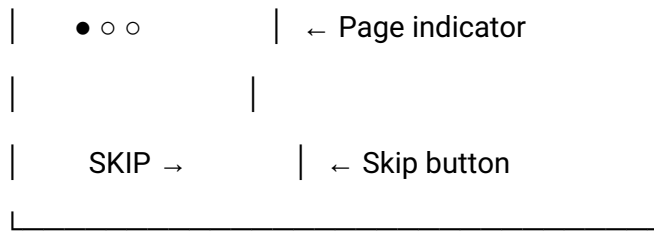
Animation:

- Background: Dark gradient (#1A1A2E → #16213E)
- Spice icon: Scale 0 → 1.2 → 1, rotation 0 → 360°, duration 1.5s
- Floating particles: 15-20 spice emoji floating upward
- Logo text: Letter-by-letter typing animation
- Loading bar: Smooth width animation 0% → 100%
- Exit: Fade out + scale down, transition to Onboarding

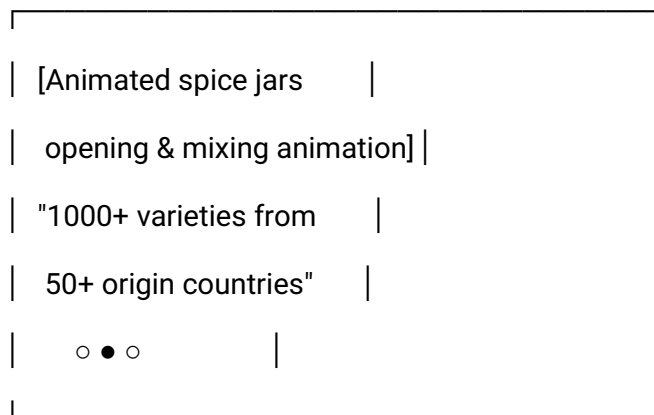
## Screen 2: ONBOARDING (3 slides, swipeable)

### SLIDE 1: "Discover Authentic Spices"

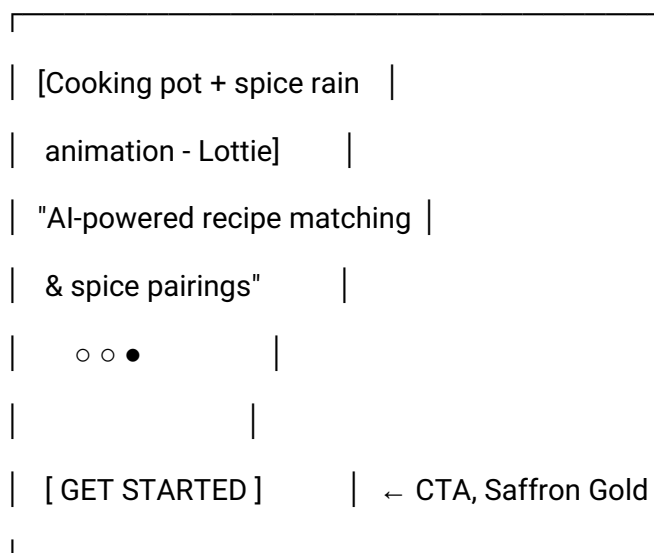




## SLIDE 2: "Every Spice, Every Blend"



## SLIDE 3: "Cook with Confidence"





Screen 3: LOGIN / SIGNUP

← Back



SpiceCart

← Brand logo

"Welcome back!"

Phone Number

← Country code + number

Password

← Show/hide toggle

LOGIN

← Saffron Gold button

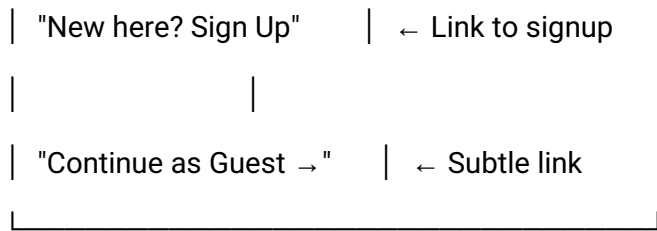
"—— or continue with ——"

G





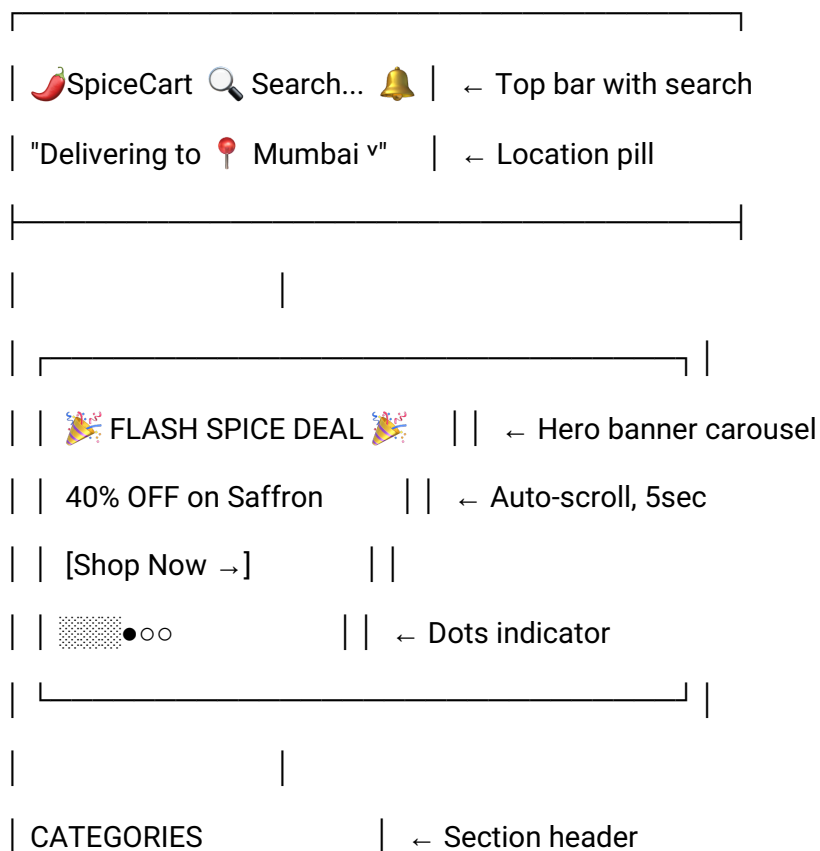
← Google, Phone OTP, Apple

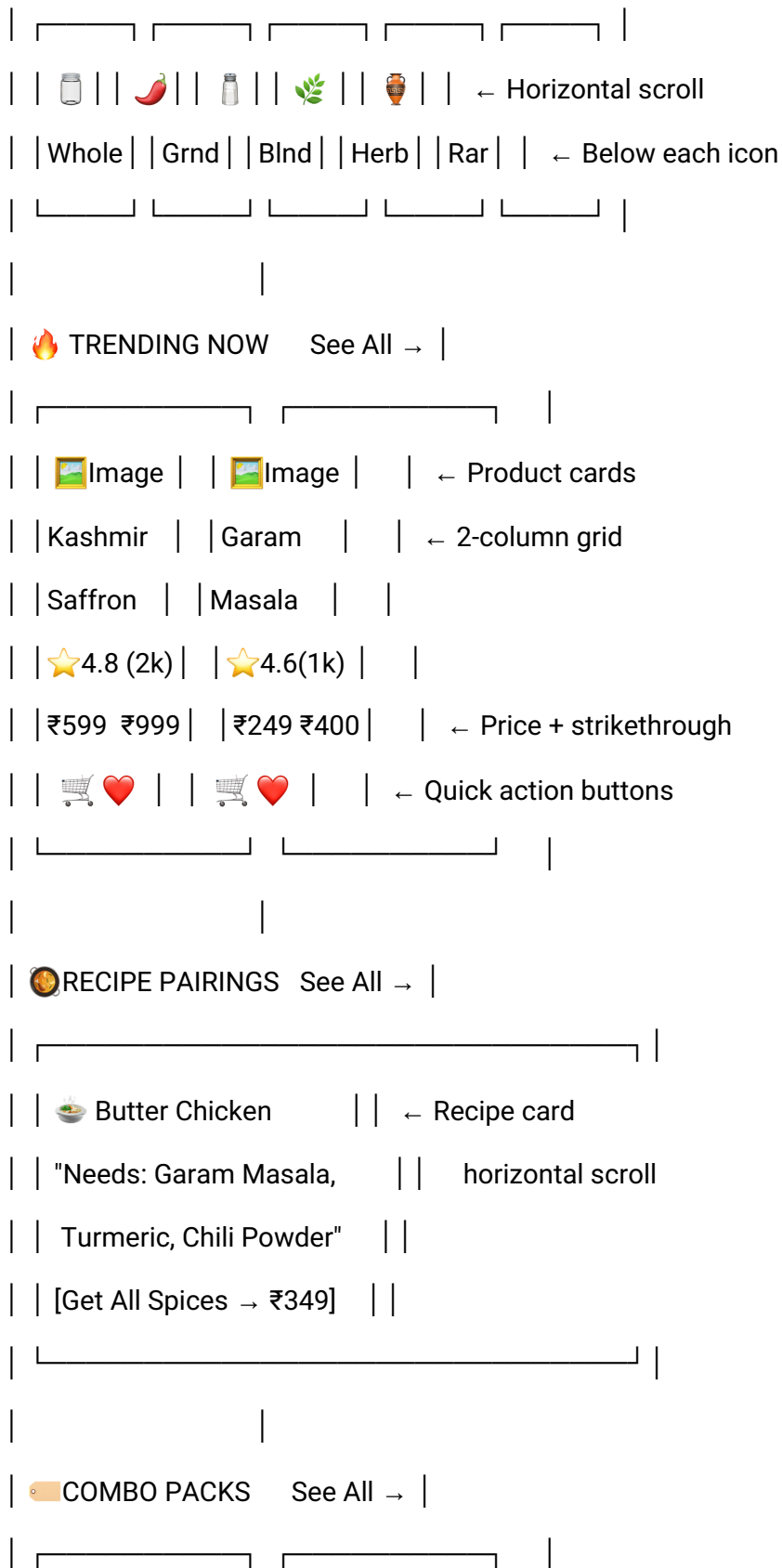


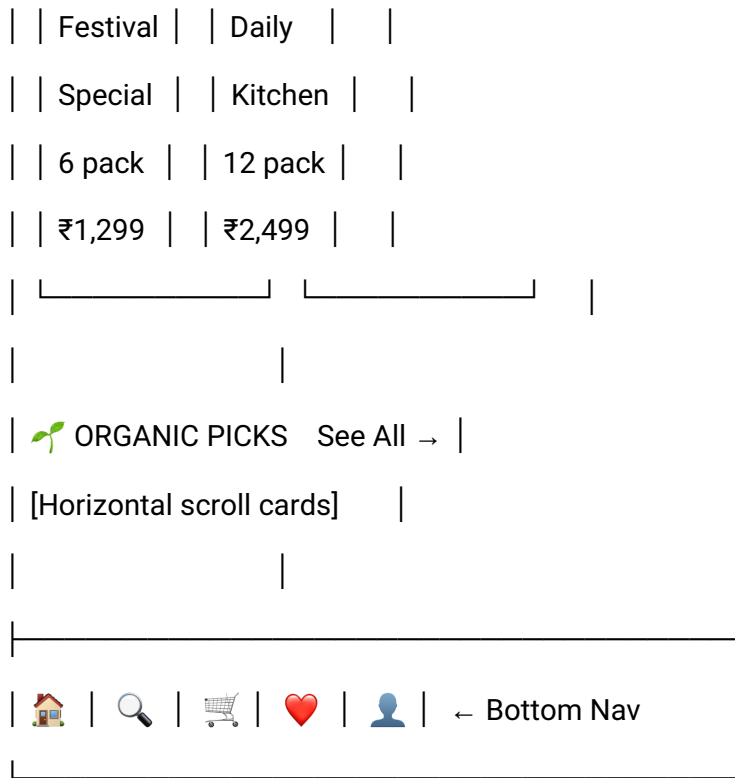
Animation:

- Background: Animated gradient mesh (amber waves)
- Login button: Shimmer effect on hover/press
- Social buttons: Scale 0.95 on press with ripple
- Input fields: Focus border animation (gold glow)
- Error: Shake animation on validation fail

Screen 4: HOME SCREEN (Main Hub — Most Complex)







Home Screen Animations:

Banner: Auto-scroll with parallax, 3D card flip on swipe

Categories: Spring animation on tap, scale 1 → 1.15 → 1

Product cards: Staggered fade-in on scroll (using IntersectionObserver)

Add to cart: Item flies to cart icon (path animation)

Wishlist: Heart fills with pulsing animation

Pull to refresh: Custom spice-jar pour animation

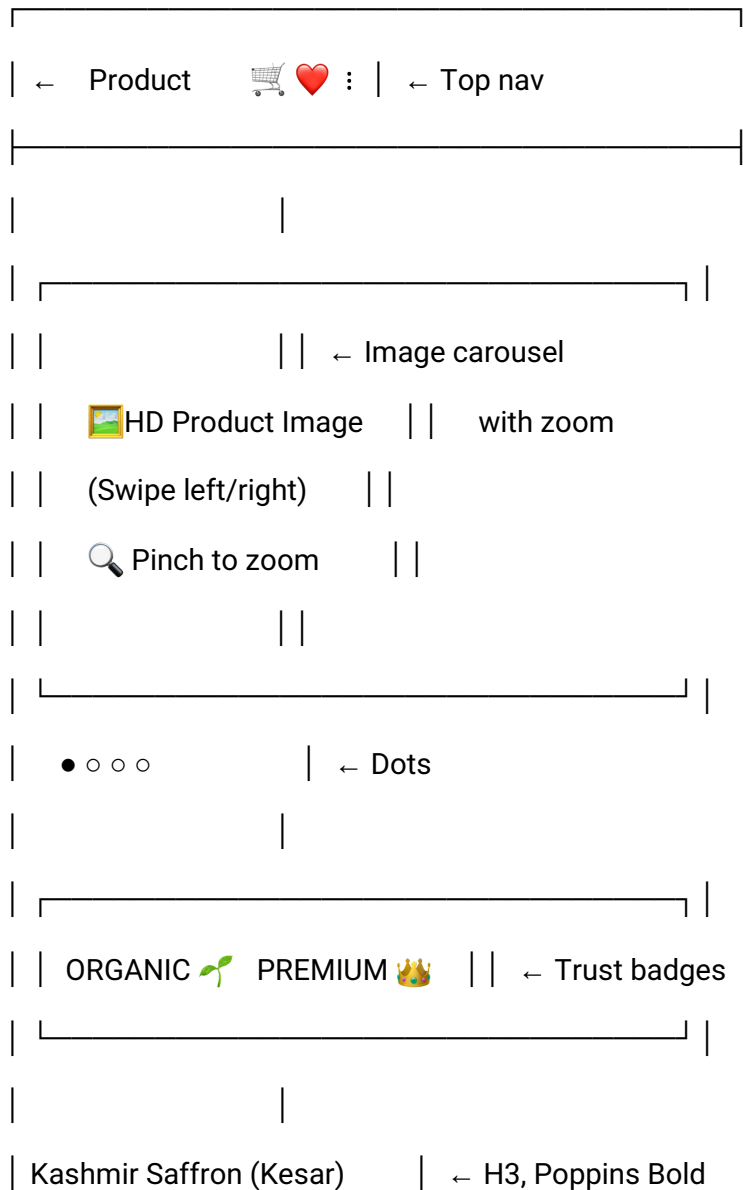
Flash deal: Countdown timer with flip-clock animation



## Animations:

- Search: Live suggestions with typing debounce
- Filter chips: Spring pop animation on select
- Grid/List toggle: Shared axis transition
- Scroll: Parallax on product images
- Loading: Skeleton shimmer animation

## Screen 6: PRODUCT DETAIL PAGE (PDP)



| by SpiceVilla | ← Brand link

| |

| ★★★★★ 4.8 (2,847 reviews) | ← Rating

| 🌶️ | ← Low heat

| |

| ₹ 599 ₹ 999 40% OFF | ← Price block

| Inclusive of all taxes |

| |

| — VARIANTS — |

| [1g] [2g] [5g] [10g ●] | ← Size selector

| |

| — ORIGIN & INFO — |

| 📍 Kashmir, India |

| 🏷️ 100% Authentic |

| 📦 Ships in 24hrs |

| 🔄 Easy Returns |

| |

| — SPICE HEAT METER — |

| 🌶️ 1/10 | ← Custom heat bar

| |

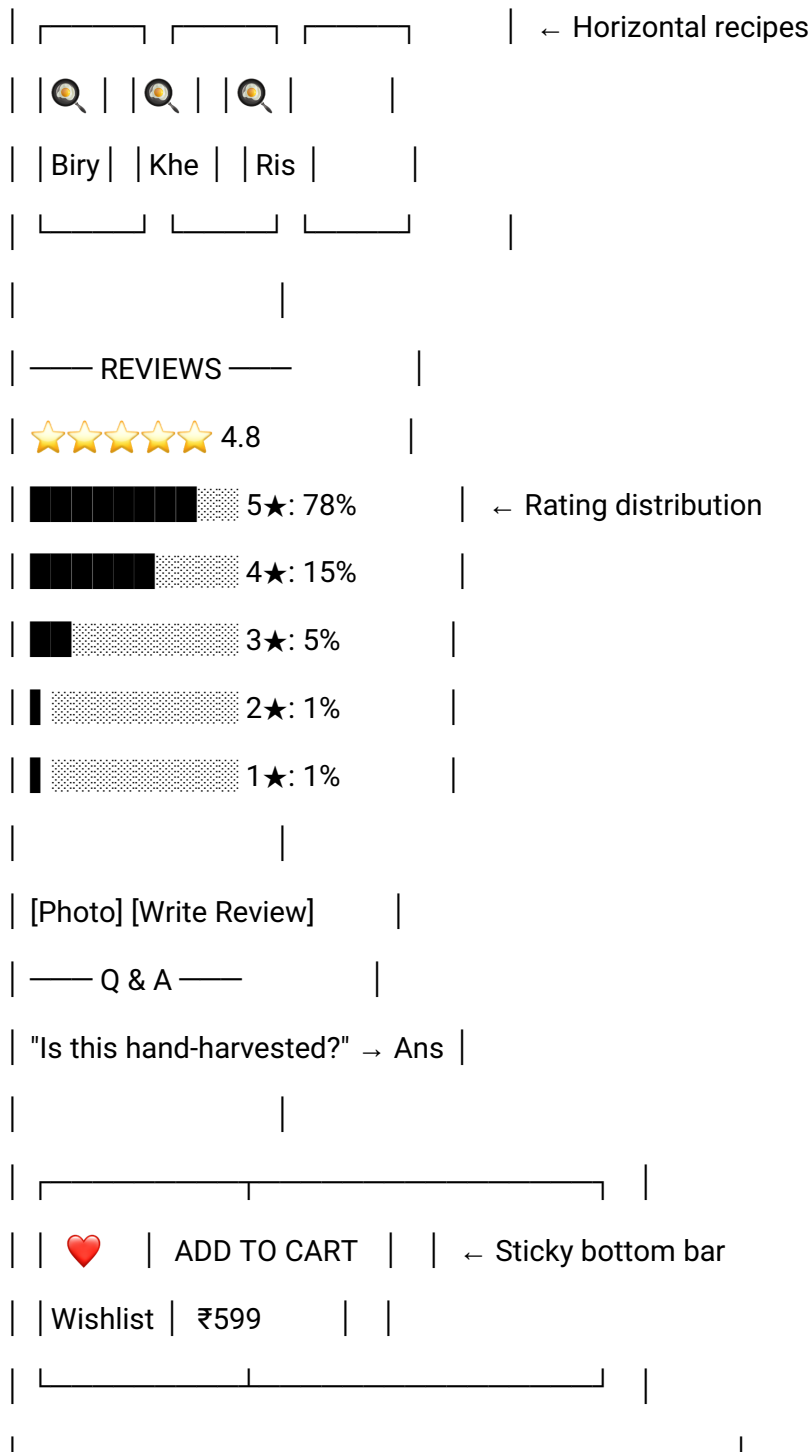
| — NUTRITION (per 100g) — |

| Calories: 310 | Fiber: 3.5g |

| Iron: 12mg | Vit A: 1500IU |

| |

| — RECIPES USING THIS — |



PDP Animations:

- Image gallery: SharedElement transition from list



- Pinch zoom: Smooth with double-tap reset
- Heat meter: Animated fill on visible
- Add to cart: Button morph + check mark
- Sticky bar: Slide up on scroll
- Trust badges: Fade in staggered
- Reviews: Expand/collapse smooth transition

## Screen 7: CART

|                                                                                               |                    |  |
|-----------------------------------------------------------------------------------------------|--------------------|--|
| ← My Cart (3 items)                                                                           |                    |  |
|                                                                                               |                    |  |
|                                                                                               |                    |  |
|                                                                                               |                    |  |
|            | Kashmir Saffron    |  |
| 10g                                                                                           | ₹599               |  |
| [-] 1 [+]  | ← Quantity stepper |  |
|                                                                                               |                    |  |
|                                                                                               |                    |  |
|            | Garam Masala       |  |
| 250g                                                                                          | ₹249               |  |
| [-] 2 [+]  |                    |  |
|                                                                                               |                    |  |
|                                                                                               |                    |  |
|            | Turmeric Powder    |  |
| 500g                                                                                          | ₹179               |  |
| [-] 1 [+]  |                    |  |

SPICE PAIRING SUGGESTION

"Add Cardamom for your Biryani recipe? +₹299"

[Add] [Dismiss]

Apply Coupon

[SPICE20 → APPLY] ← Coupon input

PRICE BREAKDOWN

Subtotal (4 items): ₹1,426

Discount (20%): -₹285

Delivery: ₹49 ← Free above ₹499

Total: ₹1,190

 PROCEED TO CHECKOUT

 ← Sticky button

Cart Animations:

- Swipe to delete: Slide left animation
- Qty change: Number spring animation
- Coupon apply: Success confetti mini
- Price update: Counter animation (₹ slides)
- Pairing suggestion: Slide-in from right
- Empty cart: Lottie animation (empty jar)

## Screen 8: CHECKOUT

← Checkout

STEP

← Step indicators

STEP 1: DELIVERY ADDRESS

📍 Home

123 Spice Lane, Mumbai

Maharashtra - 400001

Mobile: +91 98xxx xxxxx

[Edit] [+]

STEP 2: DELIVERY SLOT

Today 2-5PM

○ Tomorrow ●

17

Thu 9AM-12PM

Fri 3-6PM

← Horizontal date slots

STEP 3: PAYMENT

☐ UPI (GPay/PhonePe)

☒ Credit/Debit Card

☐ Net Banking

☐ Cash on Delivery

☐ SpiceCart Wallet (₹200)

ORDER SUMMARY

3 items | ₹1,190

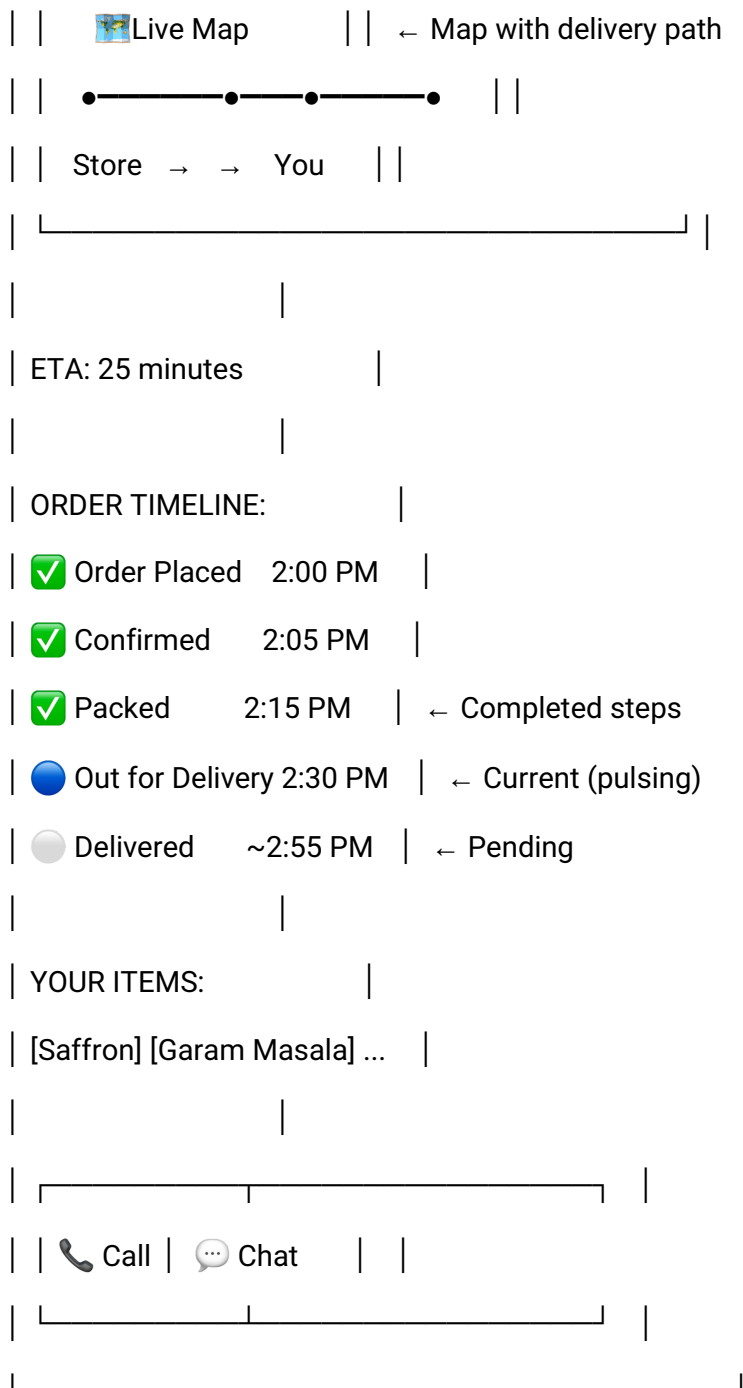
☒ I agree to Terms & Conditions

 PLACE ORDER ₹1,190

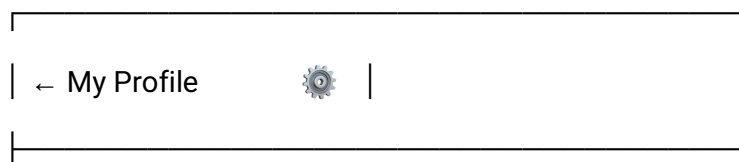
Screen 9: ORDER TRACKING

← Order #SPC-2847





## Screen 10: PROFILE



|  |                                                                                                          |                      |
|--|----------------------------------------------------------------------------------------------------------|----------------------|
|  |                                                                                                          |                      |
|  |                                                                                                          |                      |
|  |  John Doe               |                      |
|  | john@email.com                                                                                           |                      |
|  |  SpiceCraft SPICE LOVER | ← Loyalty tier badge |
|  |  1,250 Spice Points     |                      |
|  |                                                                                                          |                      |
|  |                                                                                                          |                      |
|  |  My Orders              | →                    |
|  |  Saved Addresses        | →                    |
|  |  Payment Methods        | →                    |
|  |  Wishlist             | →                    |
|  |  Spice Points         | →                    |
|  |  Notifications        | →                    |
|  |  Dark Mode            | [ON/OFF]             |
|  |  Language             | English →            |
|  |  AI Recommendations   | →                    |
|  |  Help & Support       | →                    |
|  |  About                | →                    |
|  |  Logout               |                      |
|  |                                                                                                          |                      |
|  |                                                                                                          |                      |
|  | v2.1.0                                                                                                   |                      |

## 5. 🧩 COMPONENT LIBRARY

### Core Components Table

#### Component

| Props               | Animation                           | Notes                                      |
|---------------------|-------------------------------------|--------------------------------------------|
| SpiceSearchBar      | placeholder, onSearch, voiceEnabled | Focus expand, ripple Debounced live search |
| SpiceCategoryCard   | icon, label, color, count           | Spring tap, scale Gradient backgrounds     |
| SpiceProductCard    | product, variant, layout            | Stagger fade, fly-to-cart Grid/List modes  |
| SpiceHeatMeter      | level (1-10), animated              | Animated fill Color gradient green → red   |
| SpicePriceTag       | price, originalPrice, discount      | Counter slide Smart rounding               |
| SpiceQtyStepper     | min, max, value, onChange           | Number spring Haptic feedback              |
| SpiceCouponInput    | onApply, onSuccess                  | Success confetti Auto-validate             |
| SpiceRatingBar      | rating, count, size                 | Star fill animation Half-star support      |
| SpiceCountdownTimer | endTime                             | Flip digits For flash deals                |
| SpiceEmptyState     | type, message, action               | Lottie loop Empty cart/order/search        |
| SpiceShimmerLoader  | variant (card/list/banner)          | Shimmer sweep Skeleton loading             |
| SpiceToast          | type, message, duration             | Slide in/out Success/error/info            |
| SpiceBottomSheet    | content, isOpen                     | Spring slide-up With backdrop blur         |
| SpiceCarousel       | items, autoPlay, interval           | Page transition Parallax option            |
| SpiceStepIndicator  | steps, current                      | Line animation Checkout flow               |
| SpiceMapView        | markers, route                      | Pin drop Order tracking                    |
| SpiceTrustBadge     | type, label                         | Fade stagger Organic/Premium/Auth          |
| SpiceChipFilter     | options, selected                   | Bounce select Multi-select                 |
| SpicePullToRefresh  | onRefresh                           | Jar pour LottieCustom animation            |
| SpiceFloatingCTA    | label, price, onClick               | Slide up                                   |

## 6. 🎬 ANIMATION & MICRO-INTERACTION LIBRARY

### Comprehensive Animation Catalog

#### A. Page Transitions

##### Transition

| Library | Duration | Easing |
|---------|----------|--------|
|---------|----------|--------|

|                          |               |                                    |
|--------------------------|---------------|------------------------------------|
| Shared Axis (Horizontal) | Motion Layout | 300ms cubic-bezier(0.4, 0, 0.2, 1) |
|--------------------------|---------------|------------------------------------|

|                                        |                    |                 |
|----------------------------------------|--------------------|-----------------|
| Shared Element (image from list → PDP) | Hero/SharedElement | 400ms EaseInOut |
|----------------------------------------|--------------------|-----------------|

|                            |               |               |
|----------------------------|---------------|---------------|
| Fade Through (bottom tabs) | Motion Layout | 250ms EaseOut |
|----------------------------|---------------|---------------|

|                                     |                 |                      |
|-------------------------------------|-----------------|----------------------|
| Container Transform (card → detail) | Material Motion | 350ms EaseInOutCubic |
|-------------------------------------|-----------------|----------------------|

|                                |        |                             |
|--------------------------------|--------|-----------------------------|
| Slide Up (bottom sheet, modal) | Custom | 300ms Spring (damping: 0.8) |
|--------------------------------|--------|-----------------------------|

|                           |        |               |
|---------------------------|--------|---------------|
| Slide+Fade (notification) | Custom | 200ms EaseOut |
|---------------------------|--------|---------------|

#### B. Micro-Interactions

##### Interaction

| Animation | Duration | Trigger |
|-----------|----------|---------|
|-----------|----------|---------|

|             |                                       |                  |
|-------------|---------------------------------------|------------------|
| Add to Cart | Item flies to cart icon with arc path | 600ms Button tap |
|-------------|---------------------------------------|------------------|

|          |                                    |                 |
|----------|------------------------------------|-----------------|
| Wishlist | Heart fills with pulse + particles | 400ms Heart tap |
|----------|------------------------------------|-----------------|

|              |                       |                   |
|--------------|-----------------------|-------------------|
| Quantity +/- | Number slides up/down | 200ms Stepper tap |
|--------------|-----------------------|-------------------|

|              |                                      |                     |
|--------------|--------------------------------------|---------------------|
| Swipe Delete | Item slides out left, list collapses | 300ms Swipe gesture |
|--------------|--------------------------------------|---------------------|

|              |                                       |           |
|--------------|---------------------------------------|-----------|
| Pull Refresh | Spice jar tilts, grains pour out Loop | Pull down |
|--------------|---------------------------------------|-----------|

|              |                                               |                  |
|--------------|-----------------------------------------------|------------------|
| Search Focus | Bar expands, backdrop blurs, suggestions rise | 250ms Tap search |
|--------------|-----------------------------------------------|------------------|

|               |                          |                |
|---------------|--------------------------|----------------|
| Filter Select | Chip bounces with spring | 200ms Chip tap |
|---------------|--------------------------|----------------|

|              |                         |                  |
|--------------|-------------------------|------------------|
| Rating Hover | Stars fill sequentially | 100ms/each Touch |
|--------------|-------------------------|------------------|

|              |                                 |                 |
|--------------|---------------------------------|-----------------|
| Coupon Apply | Check mark draws, badge bounces | 500ms Tap apply |
|--------------|---------------------------------|-----------------|

|              |                                     |                  |
|--------------|-------------------------------------|------------------|
| Price Change | Old price slides out, new slides in | 300ms Qty change |
|--------------|-------------------------------------|------------------|

|                 |                                    |                   |
|-----------------|------------------------------------|-------------------|
| Scroll Parallax | Product images move at 0.8x scroll | Continuous Scroll |
|-----------------|------------------------------------|-------------------|



|                    |                                     |                     |                  |
|--------------------|-------------------------------------|---------------------|------------------|
| Skeleton Shimmer   | Gradient sweeps left → right        | 1.5s loop           | Loading          |
| Button Press       | Scale 1 → 0.95 + ripple             | 150ms               | Press            |
| Success Order      | Spice confetti explosion            | 1000ms              | Order placed     |
| Error Shake        | Input field shakes left-right       | 300ms               | Validation fail  |
| Tab Switch         | Icon morphs, indicator slides       | 250ms               | Tab tap          |
| Banner Auto-scroll | 3D perspective swipe                | 500ms               | Auto 5s interval |
| Toast              | Slides up from bottom, auto-dismiss | 200ms in, 200ms out | Action result    |
| Chat Bubble        | Typing indicator, bubble pop-in     | 200ms               |                  |

### C. Lottie Animations (Required)

#### Animation

| File | Where Used | Size |
|------|------------|------|
|------|------------|------|

|                |                     |                     |        |
|----------------|---------------------|---------------------|--------|
| Spice Splash   | splash_spice.json   | Splash screen       | ~150KB |
| Empty Jar      | empty_jar.json      | Empty cart/wishlist | ~80KB  |
| Jar Pour       | jar_pour.json       | Pull to refresh     | ~60KB  |
| Spice Confetti | spice_confetti.json | Order success       | ~100KB |
| Cooking Pot    | cooking_pot.json    | Recipe pairing      | ~70KB  |
| Delivery Truck | delivery_truck.json | Order confirmation  | ~90KB  |
| Spice Mill     | spice_mill.json     | Loading states      | ~50KB  |
| Reward Star    | reward_star.json    | Points earned       | ~40KB  |
| Success Check  | success_check.json  | Payment confirmed   | ~30KB  |
| Error Pepper   | error_pepper.json   | Error states        | ~45KB  |

### D. Rive Animations (Interactive, Lightweight)

#### Animation

| State Machine | Where Used |
|---------------|------------|
|---------------|------------|

|              |                    |                         |
|--------------|--------------------|-------------------------|
| Spice Mascot | idle → wave → cook | Home floating assistant |
|--------------|--------------------|-------------------------|

Search Loader input → processing → done      Search loading

Heat Meter      level(0-10)      Product detail

Cart Badge      count(0-99)      Cart icon counter

7.  TECH STACK & ARCHITECTURE

Mobile App (Primary)

|                                                                                     |                                       |
|-------------------------------------------------------------------------------------|---------------------------------------|
| TECH STACK CHOICES                                                                  |                                       |
|                                                                                     |                                       |
|                                                                                     |                                       |
|    | FRAMEWORK: React Native (Expo SDK 50) |
|    | WEB VERSION: Next.js 14 (App Router)  |
|    | ADMIN: React + Vite                   |
|                                                                                     |                                       |
|  | LANGUAGE: TypeScript 5.3+             |
|  | STATE: Zustand + React Query          |
|  | NAV: Expo Router v3 (file-based)      |
|  | UI: NativeWind (TailwindCSS)          |
|  | ANIMATIONS: Reanimated 3 + Moti +     |
|                                                                                     | Lottie + Rive                         |
|  | IMAGES: Expo Image (BlurHash)         |
|  | AUTH: Clerk / Supabase Auth           |
|  | STORAGE: MMKV + AsyncStorage          |
|  | ANALYTICS: PostHog + Mixpanel         |
|  | CRASH: Sentry                         |
|  | PUSH: Expo Notifications + FCM        |

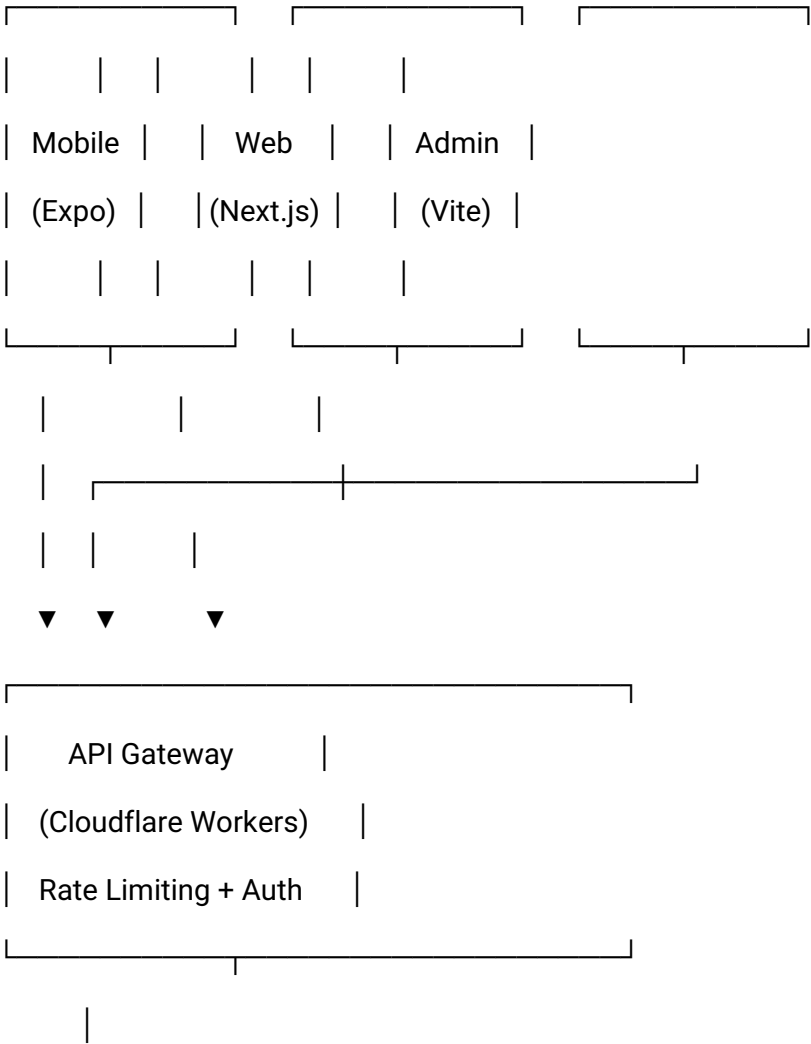
|    |          |                            |  |
|----|----------|----------------------------|--|
| 🗺️ | MAPS:    | React Native Maps          |  |
| 💳  | PAYMENT: | Razorpay / Stripe          |  |
| 🔍  | SEARCH:  | Algolia                    |  |
| 📄  | FORMS:   | React Hook Form + Zod      |  |
| 🔧  | TEST:    | Jest + Detox + Testing Lib |  |
| 📦  | CI/CD:   | EAS Build + GitHub Actions |  |
| 🤖  | E2E:     | Maestro                    |  |

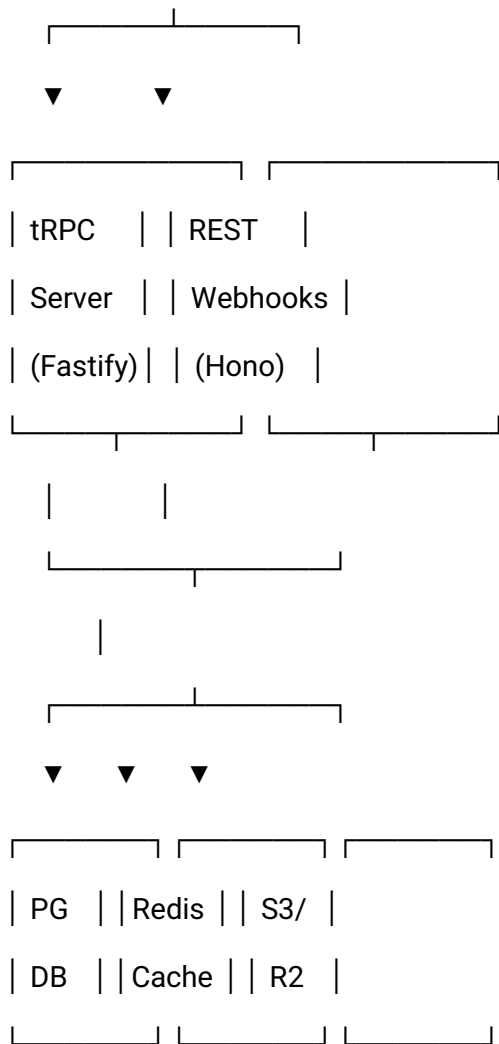
## Backend

|                      |            |                          |  |
|----------------------|------------|--------------------------|--|
| BACKEND ARCHITECTURE |            |                          |  |
|                      |            |                          |  |
|                      |            |                          |  |
| 💻                    | RUNTIME:   | Node.js 20 LTS           |  |
| 📦                    | FRAMEWORK: | Fastify 4 / Hono         |  |
| 🗄️                   | DATABASE:  | PostgreSQL 16 (Supabase) |  |
| 🔴                    | CACHE:     | Redis 7 (Upstash)        |  |
| 🔍                    | SEARCH:    | Meilisearch / Algolia    |  |
| 📁                    | STORAGE:   | Cloudflare R2 / S3       |  |
| 📧                    | QUEUE:     | BullMQ / Inngest         |  |
| 🔑                    | AUTH:      | Supabase Auth            |  |
| 🏠                    | ORM:       | Drizzle ORM              |  |
| 📡                    | REALTIME:  | Supabase Realtime        |  |
| ✉️                   | EMAIL:     | Resend                   |  |
| 📱                    | PUSH:      | Firebase Cloud Messaging |  |

|   |            |                           |
|---|------------|---------------------------|
| 📄 | PAYMENT:   | Razorpay + Stripe         |
| 🌐 | DEPLOY:    | Docker + Fly.io / Railway |
| 📈 | MONITOR:   | Grafana + Prometheus      |
| 📊 | ANALYTICS: | PostHog                   |
| 🔗 | API:       | tRPC / REST hybrid        |
| 📐 | SCHEMA:    | Zod validation            |

Architecture Diagram





## 8. 🚩 STATE MANAGEMENT & DATA FLOW

// — STORES —

// 1. Auth Store

```

interface AuthStore {
  user: User | null;
  isAuthenticated: boolean;
  login: (credentials) => Promise<void>;
}
  
```

```
logout: () => void;
guestMode: boolean;
}
```

// 2. Cart Store (persisted)

```
interface CartStore {
  items: CartItem[];
  couponCode: string | null;
  discount: number;
  addItem: (product, qty, variant) => void;
  removeItem: (productId) => void;
  updateQty: (productId, qty) => void;
  clearCart: () => void;
  applyCoupon: (code) => Promise<boolean>;
  total: number;    // computed
  itemCount: number; // computed
}
```

// 3. Wishlist Store (persisted)

```
interface WishlistStore {
  items: WishlistItem[];
  toggle: (product) => void;
  isInWishlist: (productId) => boolean;
}
```

// 4. Search & Filter Store

```
interface SearchStore {  
  query: string;  
  filters: FilterState;  
  sortBy: SortOption;  
  viewMode: 'grid' | 'list';  
  setQuery: (q) => void;  
  toggleFilter: (key, value) => void;  
  clearFilters: () => void;  
}
```

// 5. UI Store

```
interface UIStore {  
  theme: 'light' | 'dark' | 'system';  
  onboardingComplete: boolean;  
  activeTab: string;  
  isSearchOpen: boolean;  
  showToast: (type, message) => void;  
}
```

### React Query Cache Strategy

// Stale Times per Data Type

```
const STALE_TIMES = {  
  categories: 1000 * 60 * 30, // 30 minutes
```

```
products:    1000 * 60 * 5, // 5 minutes
searchResults: 1000 * 60 * 2, // 2 minutes
cart:        1000 * 30,    // 30 seconds
userProfile: 1000 * 60 * 10, // 10 minutes
orders:      1000 * 60 * 5, // 5 minutes
reviews:     1000 * 60 * 15, // 15 minutes
recommendations: 1000 * 60 * 10, // 10 minutes
flashDeals:  1000 * 10,    // 10 seconds (dynamic!)
};
```

## 9. 📡 API DESIGN & BACKEND

### REST + tRPC Endpoints

#### AUTH

- |— POST /auth/phone/send-otp
- |— POST /auth/phone/verify-otp
- |— POST /auth/email/login
- |— POST /auth/email/register
- |— POST /auth/refresh
- |— POST /auth/logout

#### PRODUCTS

- |— GET /products → List (paginated, filtered)
- |— GET /products/:id → Detail
- |— GET /products/:id/variants → Variant options
- |— GET /products/:id/reviews → Reviews (paginated)



- └─ POST /products/:id/reviews → Create review
- └─ GET /products/:id/related → Similar products
- └─ GET /products/:id/recipes → Paired recipes
- └─ GET /products/search → Search (Algolia)

## CATEGORIES

- └─ GET /categories → All categories
- └─ GET /categories/:slug/products → Category products
- └─ GET /categories/tree → Hierarchical tree

## CART

- └─ GET /cart → Get cart
- └─ POST /cart/items → Add item
- └─ PATCH /cart/items/:id → Update quantity
- └─ DELETE /cart/items/:id → Remove item
- └─ POST /cart/coupon → Apply coupon
- └─ DELETE /cart/coupon → Remove coupon

## ORDERS

- └─ GET /orders → List orders
- └─ GET /orders/:id → Order detail
- └─ POST /orders → Create order
- └─ POST /orders/:id/cancel → Cancel order
- └─ GET /orders/:id/track → Track order

## WISHLIST

- └─ GET /wishlist → List items
- └─ POST /wishlist/:productId → Add
- └─ DELETE /wishlist/:productId → Remove

## USER

- └─ GET /user/profile → Profile info
- └─ PATCH /user/profile → Update profile
- └─ GET /user/addresses → List addresses
- └─ POST /user/addresses → Add address
- └─ PATCH /user/addresses/:id → Update address
- └─ DELETE /user/addresses/:id → Delete address
- └─ GET /user/points → Spice points

## PAYMENT

- └─ POST /payment/create-order → Razorpay order
- └─ POST /payment/verify → Verify payment
- └─ POST /payment/webhook → Razorpay webhook
- └─ GET /payment/methods → Saved methods

## ADMIN

- └─ GET /admin/dashboard → Stats
- └─ CRUD /admin/products → Product management
- └─ CRUD /admin/categories → Category management
- └─ CRUD /admin/orders → Order management

|— CRUD /admin/coupons → Coupon management  
|— CRUD /admin/users → User management  
|— POST /admin/banners → Banner management  
|— GET /admin/analytics → Analytics

Database Schema (PostgreSQL + Drizzle)

-- CORE TABLES

```
users (  
  id, email, phone, name, avatar_url,  
  loyalty_points, tier, created_at, updated_at  
)
```

```
addresses (  
  id, user_id, line1, line2, city, state,  
  pincode, country, is_default, type('home'|'work')  
)
```

```
categories (  
  id, name, slug, icon_url, color_hex,  
  parent_id, display_order, is_active  
)
```

```
products (  
  id, name, slug, icon_url, color_hex,  
  parent_id, display_order, is_active  
)
```

```
id, name, slug, description, short_description,  
category_id, brand_id, origin_country, region,  
spice_heat_level, is_organic, is_premium,  
nutrition_info(json), storage_tips,  
images(jsonb), search_vector(tsvector),  
avg_rating, review_count, total_sold,  
is_active, created_at, updated_at  
)
```

```
product_variants (  
id, product_id, weight_value, weight_unit,  
mrp, selling_price, stock_quantity,  
sku, is_default  
)
```

```
brands (  
id, name, slug, logo_url, description, is_verified  
)
```

```
carts (  
id, user_id, coupon_id, created_at, updated_at  
)
```

```
cart_items (  
id, cart_id, product_id, variant_id,
```

```
    quantity, price_at_addition  
  )
```

```
orders (  
  id, user_id, order_number, status,  
  subtotal, discount, delivery_fee, total,  
  coupon_id, address_snapshot(jsonb),  
  payment_method, payment_id,  
  tracking_id, delivery_partner,  
  estimated_delivery, delivered_at,  
  created_at  
)
```

```
order_items (  
  id, order_id, product_id, variant_id,  
  quantity, price, product_snapshot(jsonb)  
)
```

```
reviews (  
  id, user_id, product_id, rating, title,  
  body, images(jsonb), is_verified_purchase,  
  helpful_count, created_at  
)
```

```
wishlists (  

```

```
id, user_id, product_id, created_at  
)
```

```
coupons (  
  id, code, type('flat'|'percent'), value,  
  min_order, max_discount, usage_limit,  
  usage_count, valid_from, valid_until, is_active  
)
```

```
recipes (  
  id, name, slug, image_url, prep_time,  
  cook_time, servings, cuisine_type,  
  ingredients(jsonb), steps(jsonb),  
  spice_pairings(jsonb)  
)
```

```
flash_deals (  
  id, product_id, variant_id, discount_percent,  
  start_time, end_time, max_units, sold_units,  
  is_active  
)
```

```
notifications (  
  id, user_id, type, title, body, data(jsonb),  
  is_read, created_at
```

)

## 10. ACCESSIBILITY & PERFORMANCE

Accessibility (WCAG 2.1 AA)

Feature

Implementation

Semantic HTML      Proper heading hierarchy, landmarks, roles

Screen Reader accessibilityLabel on all interactive elements

Color Contrast All text  $\geq 4.5:1$  ratio (verified with palette)

Touch Targets Minimum 44x44pt all buttons

Focus Order    Logical tab order, focus trap on modals

Text Scaling    Support up to 200% without layout break

Reduced Motion      useReducedMotion() disables animations

Color Blind      Never rely on color alone (icons + text)

Announcements      Live regions for cart updates, toasts

Performance Targets

Metric

Target Strategy

App Size      <25MB Optimize assets, lazy load

Time to Interactive    <2s    Splash → home quick load

First Content Paint    <1.5s    Server components, streaming

Image Load    BlurHash + progressive      WebP/AVIF, CDN

API Response <200ms      Redis cache, DB indexing

List Scroll      60fps    FlashList, window rendering

Animation      60fps    Worklet thread (Reanimated)

Bundle Split    Route-based    Dynamic imports

Memory      <200MB      Image recycling, list virtualization

### Image Optimization Strategy

- Format: AVIF (primary) > WebP > JPEG (fallback)
- CDN: Cloudflare Images (auto-transform)
- Lazy Load: IntersectionObserver + BlurHash placeholder
- Responsive: srcset with 1x, 2x, 3x variants
- Thumbnails: 200x200, compressed q70
- Product Images: 800x800, compressed q80
- Zoom: Full-res 2000x2000 loaded on demand
- Cache: HTTP Cache-Control, 30-day immutable

## 11. 📁 FULL FILE STRUCTURE

SpiceCart/

```
├── 📁 apps/
│   ├── mobile/                # React Native (Expo)
│   │   ├── app/              # Expo Router (file-based routing)
│   │   │   ├── _layout.tsx    # Root layout (providers, theme)
│   │   │   ├── index.tsx      # Splash → Onboarding redirect
│   │   │   └── (auth)/
│   │   │       ├── _layout.tsx # Auth stack layout
│   │   │       ├── login.tsx
│   │   │       ├── signup.tsx
│   │   │       └── otp-verify.tsx
```



```
| | | | └─ forgot-password.tsx
| | | └─ (tabs)/
| | | | └─ _layout.tsx      # Bottom tab navigation
| | | | └─ index.tsx       # Home screen
| | | | └─ explore.tsx     # Explore/Search
| | | | └─ cart.tsx
| | | | └─ wishlist.tsx
| | | | └─ profile.tsx
| | | └─ product/
| | | | └─ [id].tsx        # Product detail (dynamic)
| | | └─ category/
| | | | └─ [slug].tsx      # Category listing
| | | └─ checkout/
| | | | └─ index.tsx      # Checkout flow
| | | | └─ payment.tsx
| | | | └─ success.tsx
| | | └─ order/
| | | | └─ index.tsx      # Order list
| | | | └─ [id].tsx       # Order detail + tracking
| | | └─ recipe/
| | | | └─ [slug].tsx     # Recipe detail
| | | └─ search.tsx      # Full search page
| | | └─ notifications.tsx
| | |
| | └─ components/
```

```
| | | └─ ui/          # Primitive UI components
| | | | └─ Button.tsx
| | | | └─ Input.tsx
| | | | └─ Card.tsx
| | | | └─ Badge.tsx
| | | | └─ Chip.tsx
| | | | └─ Avatar.tsx
| | | | └─ BottomSheet.tsx
| | | | └─ Toast.tsx
| | | | └─ Modal.tsx
| | | | └─ Skeleton.tsx
| | | | └─ Stepper.tsx
| | | | └─ Switch.tsx
| | | | └─ Divider.tsx
| | | |
| | | └─ spice/       # Domain-specific components
| | | | └─ SpiceProductCard.tsx
| | | | └─ SpiceCategoryCard.tsx
| | | | └─ SpiceHeatMeter.tsx
| | | | └─ SpicePriceTag.tsx
| | | | └─ SpiceRatingBar.tsx
| | | | └─ SpiceCountdownTimer.tsx
| | | | └─ SpiceCouponInput.tsx
| | | | └─ SpiceSearchBar.tsx
| | | | └─ SpiceTrustBadge.tsx
```

- | | | | └─ SpiceEmptyState.tsx
- | | | | └─ SpiceShimmerLoader.tsx
- | | | | └─ SpiceCarousel.tsx
- | | | | └─ SpiceStepIndicator.tsx
- | | | | └─ SpiceChipFilter.tsx
- | | | | └─ SpicePullToRefresh.tsx
- | | | | └─ SpiceFloatingCTA.tsx
- | | | | └─ SpiceMapView.tsx
- | | | |
- | | | └─ layout/
- | | | | └─ ScreenWrapper.tsx
- | | | | └─ KeyboardAvoiding.tsx
- | | | | └─ StickyHeader.tsx
- | | | |
- | | | └─ animations/
- | | | | └─ FlyToCartAnimation.tsx
- | | | | └─ HeartFillAnimation.tsx
- | | | | └─ ConfettiEffect.tsx
- | | | | └─ StaggerFadeIn.tsx
- | | | | └─ NumberSpring.tsx
- | | | | └─ SharedElementTransition.tsx
- | | | |
- | | └─ stores/ # Zustand stores
- | | | └─ authStore.ts
- | | | └─ cartStore.ts

```
| | | └─ wishlistStore.ts
| | | └─ searchStore.ts
| | | └─ uiStore.ts
| | |
| | | └─ hooks/           # Custom hooks
| | | └─ useProducts.ts   # React Query hooks
| | | └─ useCategories.ts
| | | └─ useCart.ts
| | | └─ useAuth.ts
| | | └─ useSearch.ts
| | | └─ useDebounce.ts
| | | └─ useReducedMotion.ts
| | | └─ useTheme.ts
| | | └─ useHapticFeedback.ts
| | |
| | | └─ lib/
| | | └─ api/             # API client
| | | | └─ client.ts      # Axios instance + interceptors
| | | | └─ products.ts
| | | | └─ auth.ts
| | | | └─ cart.ts
| | | | └─ orders.ts
| | | | └─ user.ts
| | | └─ trpc/
| | | | └─ client.ts      # tRPC React client
```

- | | | |─ algolia/
- | | | | | └─ searchClient.ts
- | | | |─ analytics/
- | | | | | └─ posthog.ts
- | | | | | └─ events.ts
- | | | |─ payments/
- | | | | | └─ razorpay.ts
- | | | |─ utils/
- | | | | | └─ formatCurrency.ts
- | | | | | └─ calculateDiscount.ts
- | | | | | └─ spiceHeatColor.ts
- | | | | | └─ validators.ts
- | | | | | └─ constants.ts
- | | |
- | | |─ theme/
- | | | |─ colors.ts
- | | | |─ typography.ts
- | | | |─ spacing.ts
- | | | |─ shadows.ts
- | | | |─ index.ts
- | | |
- | | |─ assets/
- | | | |─ lottie/
- | | | | | └─ splash\_spice.json
- | | | | | └─ empty\_jar.json

```
| | | | └─ jar_pour.json
| | | | └─ spice_confetti.json
| | | | └─ cooking_pot.json
| | | | └─ success_check.json
| | | └─ rive/
| | | | └─ spice_mascot.riv
| | | | └─ heat_meter.riv
| | | └─ images/          # Static images
| | | └─ fonts/          # Custom fonts
| | |
| | └─ app.json
| | └─ eas.json
| | └─ tsconfig.json
| | └─ babel.config.js
| | └─ metro.config.js
| | └─ package.json
| |
| └─ web/                # Next.js 14 Web App
| | └─ app/
| | | └─ layout.tsx
| | | └─ page.tsx        # Home
| | | └─ products/
| | | | └─ page.tsx      # Listing
| | | | └─ [id]/page.tsx # Detail
| | | └─ cart/
```

```
| | | └─ checkout/
| | | └─ account/
| | | └─ api/          # Route handlers
| | └─ components/    # Shared + web-specific
| | └─ lib/
| └─ package.json
|
| └─ admin/           # Admin Dashboard
|   └─ src/
|     └─ pages/
|       └─ Dashboard.tsx
|       └─ Products.tsx
|       └─ Orders.tsx
|       └─ Customers.tsx
|       └─ Coupons.tsx
|       └─ Analytics.tsx
|       └─ Settings.tsx
|     └─ components/
|     └─ lib/
|     └─ package.json
|
└─ 📁 packages/
  └─ shared/          # Monorepo shared code
    └─ types/         # TypeScript types
      └─ product.ts
```

```
| | | └─ cart.ts
| | | └─ order.ts
| | | └─ user.ts
| | | └─ index.ts
| | └─ validators/      # Zod schemas (shared)
| | | └─ product.ts
| | | └─ auth.ts
| | | └─ checkout.ts
| | └─ constants/      # Shared constants
| | | └─ categories.ts
| | | └─ countries.ts
| | | └─ currencies.ts
| | └─ utils/          # Shared utilities
| | | └─ format.ts
| | | └─ date.ts
| | | └─ math.ts
| |
| └─ ui/               # Shared UI primitives
| | └─ src/
| | | └─ Button.tsx
| | | └─ Input.tsx
| | | └─ Badge.tsx
| | | └─ index.ts
| | └─ package.json
|
```



```
|— 🛠️ server/          # Backend
|  |— src/
|  |  |— index.ts      # Entry point
|  |  |— config/
|  |  |  |— env.ts
|  |  |  |— database.ts
|  |  |  └─ redis.ts
|  |  |— db/
|  |  |  |— schema/    # Drizzle schema
|  |  |  |  |— users.ts
|  |  |  |  |— products.ts
|  |  |  |  |— orders.ts
|  |  |  |  |— carts.ts
|  |  |  |  |— reviews.ts
|  |  |  |  └─ index.ts
|  |  |  |— migrations/
|  |  |  └─ seed.ts
|  |  |
|  |  |— trpc/
|  |  |  |— router.ts   # Root router
|  |  |  |— context.ts  # tRPC context
|  |  |  └─ routers/
|  |  |    |— auth.ts
|  |  |    |— products.ts
|  |  |    └─ cart.ts
```

```
| | | |— orders.ts
| | | |— reviews.ts
| | | |— wishlist.ts
| | | |— search.ts
| | | |— admin.ts
| | |
| | |— rest/
| | | |— routes/
| | | | |— webhooks.ts      # Razorpay webhook
| | | | |— health.ts
| | | |— middleware/
| | | |— auth.ts
| | | |— rateLimit.ts
| | | |— cors.ts
| | |
| | |— services/          # Business logic
| | | |— authService.ts
| | | |— productService.ts
| | | |— cartService.ts
| | | |— orderService.ts
| | | |— paymentService.ts
| | | |— searchService.ts
| | | |— notificationService.ts
| | | |— recommendationService.ts
| | |
```

```
| | |─ workers/          # Background jobs
| | |  |─ orderConfirmation.ts
| | |  |─ priceAlert.ts
| | |  |─ abandonedCart.ts
| | |  |─ reviewReminder.ts
| | |  └─ analyticsAggregator.ts
| | |
| | └─ utils/
| |   |─ couponEngine.ts
| |   |─ stockManager.ts
| |   └─ priceCalculator.ts
| |
| |
| └─ Dockerfile
| └─ docker-compose.yml
| └─ drizzle.config.ts
| └─ tsconfig.json
| └─ package.json
|
└─ 🗄️ supabase/          # Supabase configuration
  | └─ config.toml
  | └─ migrations/      # SQL migrations
  |   └─ seed.sql
  |
  └─ 📦 turbo/          # Turborepo config
    └─ turbo.json
```

```

|
|— .github/
|   └─ workflows/
|       └─ ci.yml
|       └─ deploy-mobile.yml
|       └─ deploy-web.yml
|       └─ deploy-server.yml
|
|— turbo.json          # Turborepo pipeline
|— package.json        # Monorepo root
|— pnpm-workspace.yaml
|— .env.example
|— .gitignore
|— README.md

```

## 12. 📅 DEVELOPMENT ROADMAP

### Phase

Duration      Deliverables

Phase 1: Foundation    Week 1-2      Monorepo setup, design system, auth, navigation skeleton

Phase 2: Core Shopping    Week 3-5      Product catalog, PDP, cart, checkout, basic search

Phase 3: Spice Features    Week 6-7      Heat meter, origin info, recipe pairing, AI recommendations

Phase 4: Rich Features    Week 8-9      Wishlist, reviews, notifications, order tracking, loyalty

Phase 5: Admin            Week 10-11    Admin dashboard, product/order management, analytics

Phase 6: Polish            Week 12-13    Animations, performance, accessibility, dark mode

|                  |            |                                               |
|------------------|------------|-----------------------------------------------|
| Phase 7: Testing | Week 14-15 | E2E tests, load testing, security audit       |
| Phase 8: Launch  | Week 16    | App store submission, soft launch, monitoring |

## KEY METRICS TO TRACK

Metric

Tool    Target

|                    |          |           |
|--------------------|----------|-----------|
| DAU/MAU            | PostHog  | 40% ratio |
| Conversion Rate    | Mixpanel | >3.5%     |
| Cart Abandonment   | Custom   | <65%      |
| Avg Order Value    | Stripe   | ₹800+     |
| App Crash Rate     | Sentry   | <0.5%     |
| App Store Rating   | Manual   | >4.5★     |
| Search-to-Purchase | PostHog  | >25%      |
| Push Open Rate     | Firebase | >15%      |